

Network Basics

What is Ip

- Internet Protocol
- Identifiy Every User On the Internet
- Consists of 4 parts with range between 1-255
- 192.168.1.1
- 176.16.13.199

Ports

- Every Ip Has large Number Of ports (65535)
- Every Port Runs a Service
- Ports Can be open , Closed , Filtered

Port example



RED
nexus



Port example

- Every Window In the building runs a service
- Ex: copy paper , extract identity Card , extract passport etc...
- You can exploit window if you know the man takes money as bribe
- You can exploit window if you know somebody related to you

Port Exploitation

- Discover The Open Port
- Know The Service Running
- Know the Version of service
- Exploit it

Domain Name

- Domain is another identifier to get the ip address
- Ex: The phone Number is hard to remember but you can remember the word (Ali , My bro etc...)
- Domain example :
- google.com
- rednexus.org
- hossamshady.net

Domain Name

- Every Domain has only one Ip
- But every Ip can handle many domains

Domain and Ip



DNS (Domain Name Server)

- To put domain over ip you need server called DNS

Domain name	ip
google.com	142.250.180.174
hossamshady1.github.io	185.199.108.153
facebook.com	102.132.97.35
rednexus.com	201.144.142.10
Hackerone.com	104.18.36.214

CIDR

CIDR : means range of IPs not only one ip
Ex: 192.158.1.0/24

Can be calculated Online or you
can calculate it by

$$2^{(32 - 24)} - 2$$

Gives you the number of IPs in CIDR

ASN

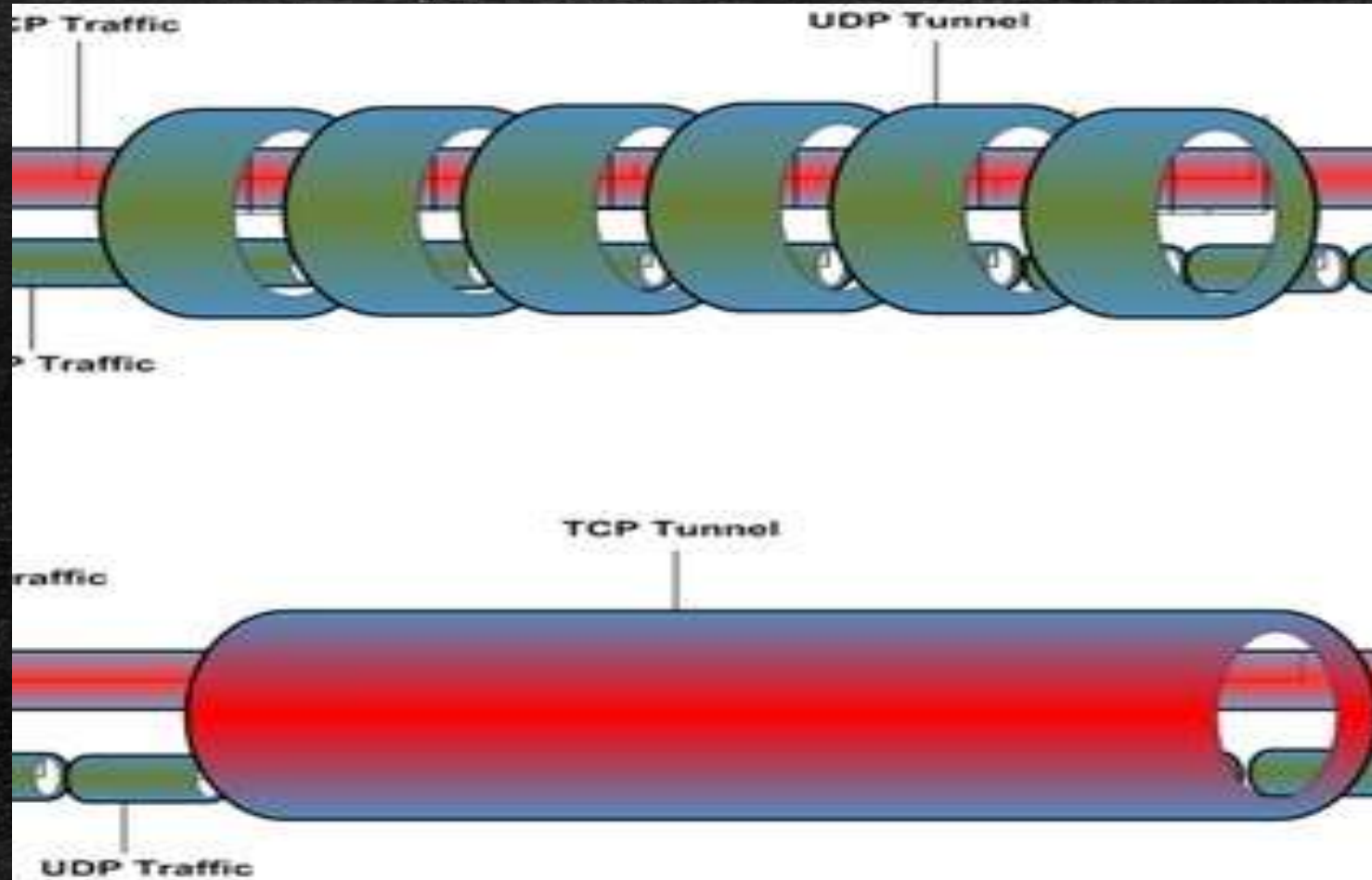
ASN (Autonomous System Number)

Identifier For Large Companies with many branches
Identify Related Ips , domains , servers , Apks

Ex: AS3303
For Swisscom LTD

Ex: AS15169
For Google

Udp and Tcp



Udp and Tcp

Encapsulation for data by adding new layers for it

Udp : Fast but not secure (Video calls , Phone calls ...

Tcp : Secure but slower (Sensitive Data , messages ...

Ex: order need to reach fast and the other need to reach secure and complete

Protocols

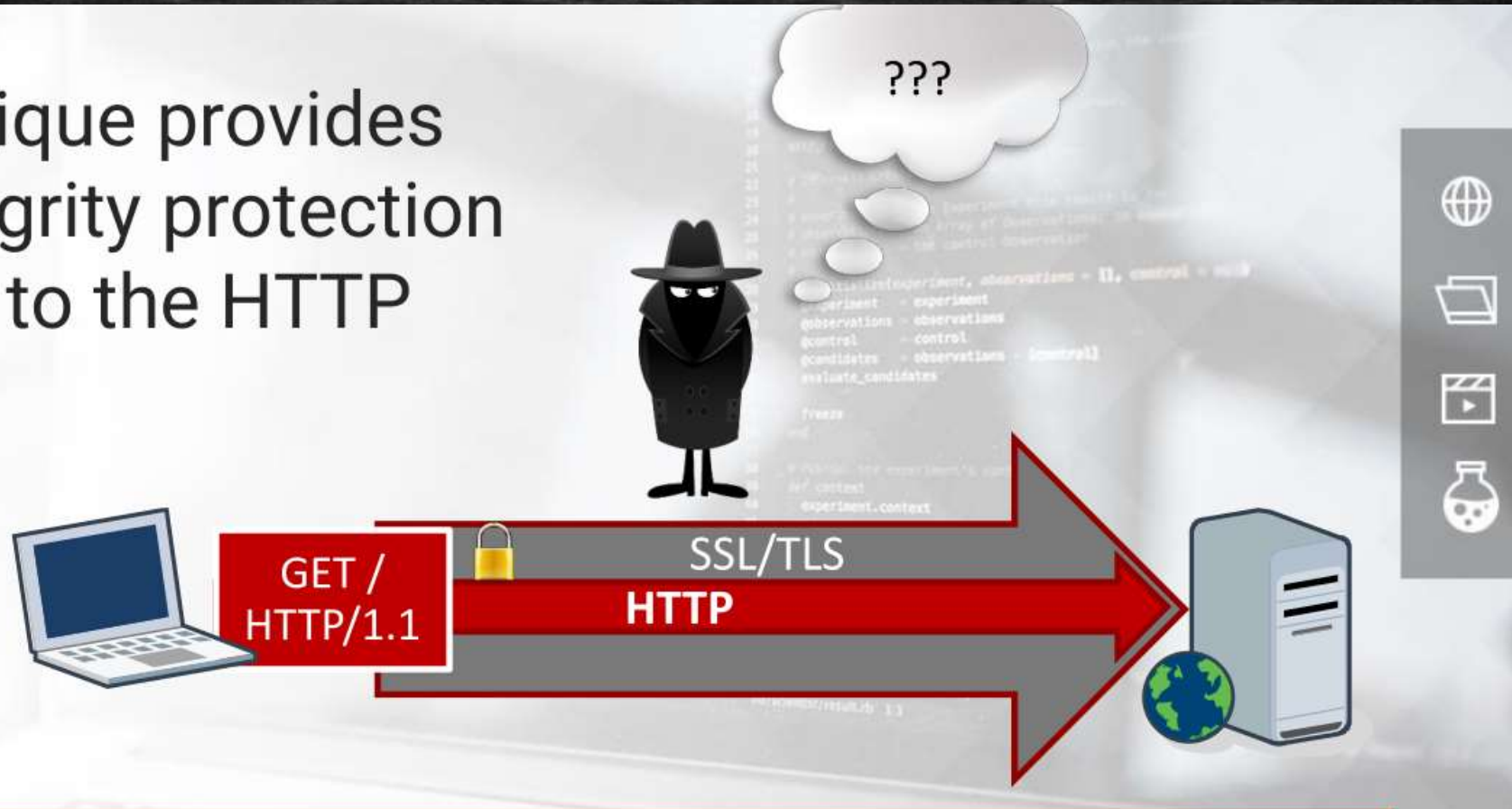
- Protocol is collection of algorithms to transfer data
- Verify there is match between two sides
- Ex:
- Http , ftp , ssh , smtp ...

SSL & TLS

- Put Certificate secure layer over data to protect it from attackers
- Added for protocol http , ftp ...
- https , sftp ...
- Encrypt the data and then put it into tunnel

SSL , TLS

Technique provides
integrity protection
to the HTTP



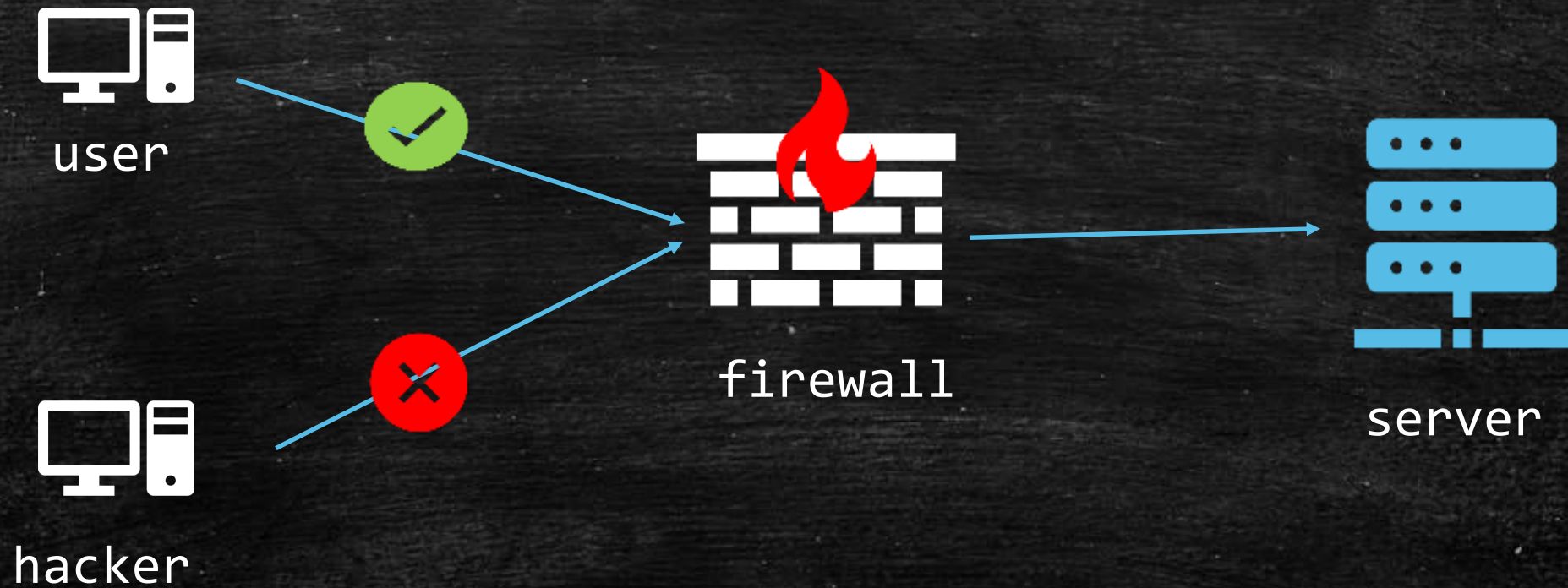
How to Contact Ali

- My private IP goes to NAT protocol
- NAT protocol converts Private IP to Public IP
- For example :
- 192.168.1.1 becomes 167.158.44.149

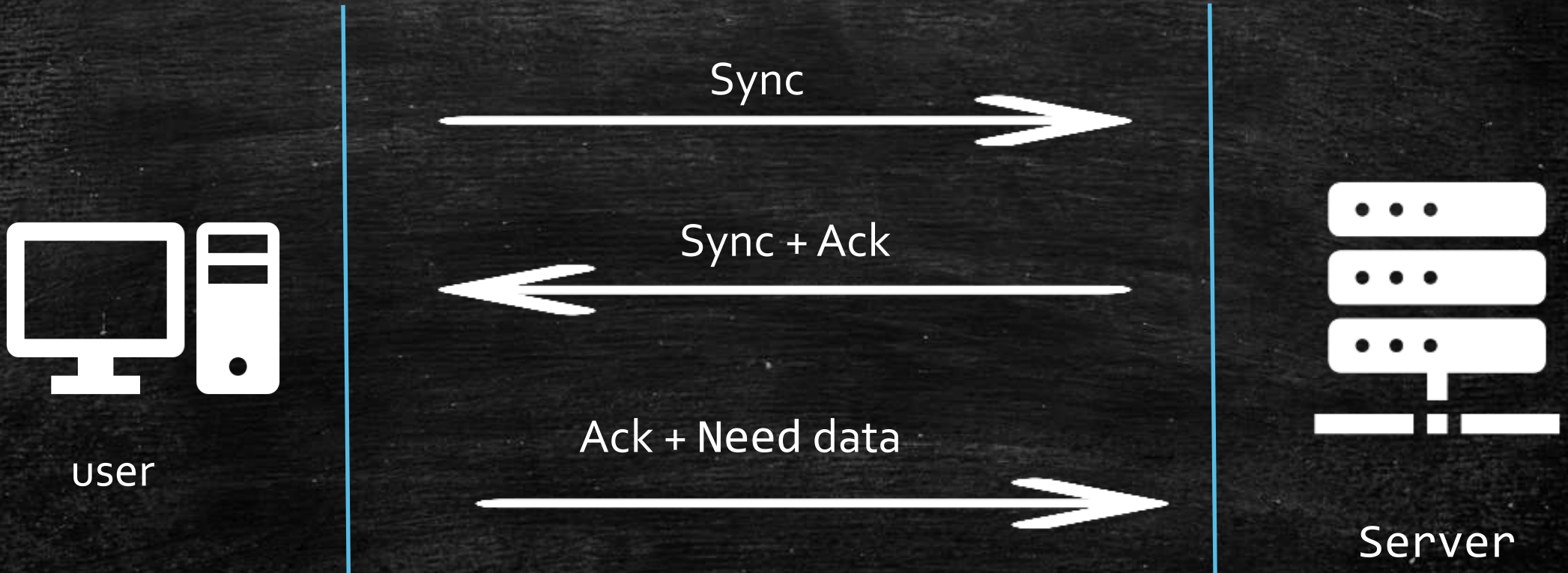
Firewall

- Firewall is device to protect company or network from attacks
- Firewall Blocks suspicious trials to enter the network and block viruses
- Firewall example :
- Akamai , Cloudflare , amazon ...

Firewall



3 Way Handshake



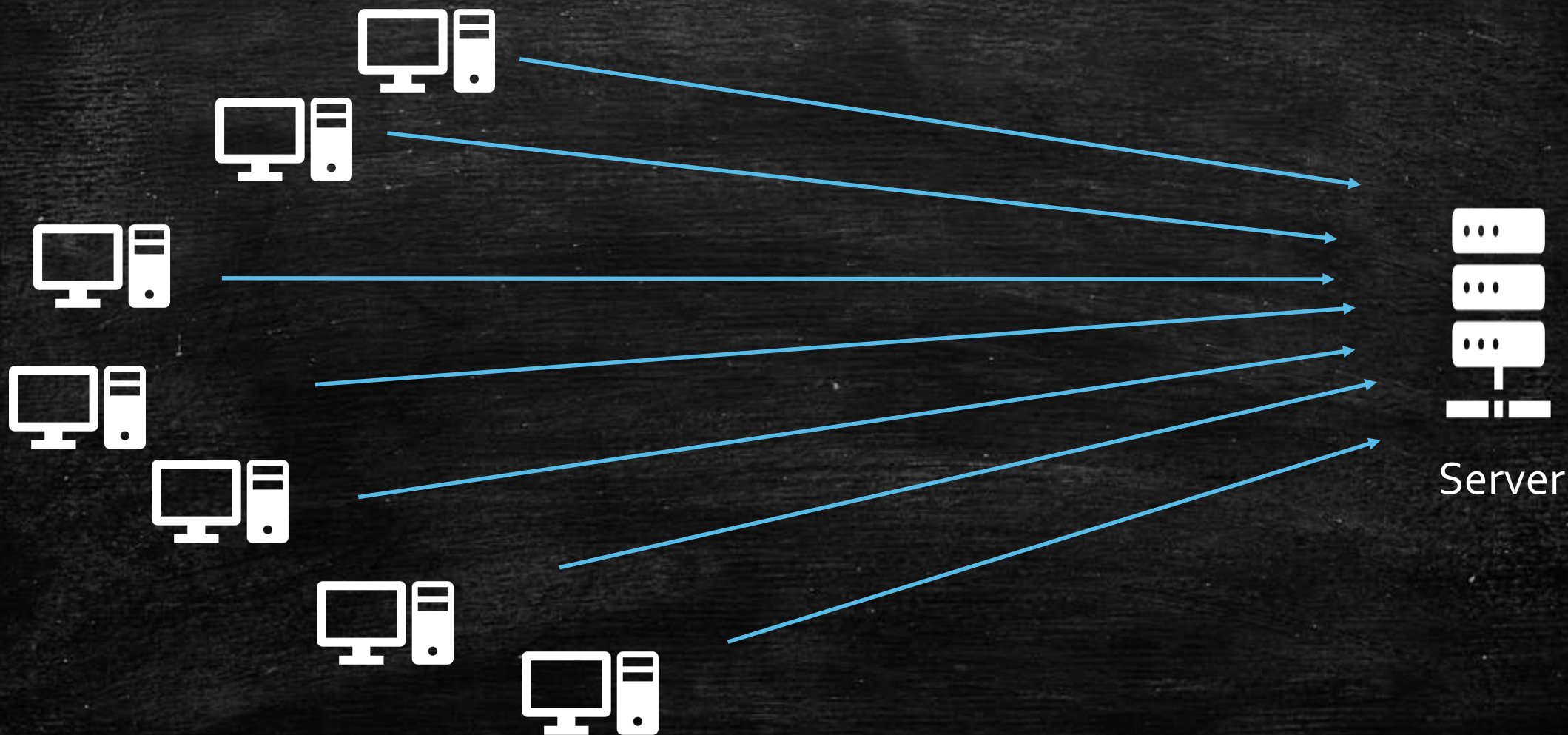
Dos Attack

- Denial of service
- It happens when large number of request are made over the server at the same time which makes the server crash
- Just add when you open multiple programs on your mobile or laptop
- Dos Prevents users from reaching to server

Dos Attack

- Suppose server can only accept 10 requests at the same time
- Attacker made 100 request at the same time
- Server can't response and then crashes and stops

Dos Attack



DDos Attack

- Same as Dos attack but with multiple devices
- The largest DDos attack was again
- 2017 with 2.54 Tb/s

What is VPS

- Virtual Private Server
- As your private PC with facilities , Rams , CPU etc...
- Many Providers as Azure , Google , Segfault
- Type :
- # ssh root@segfault.net
- And the password : segfault

Virtualization

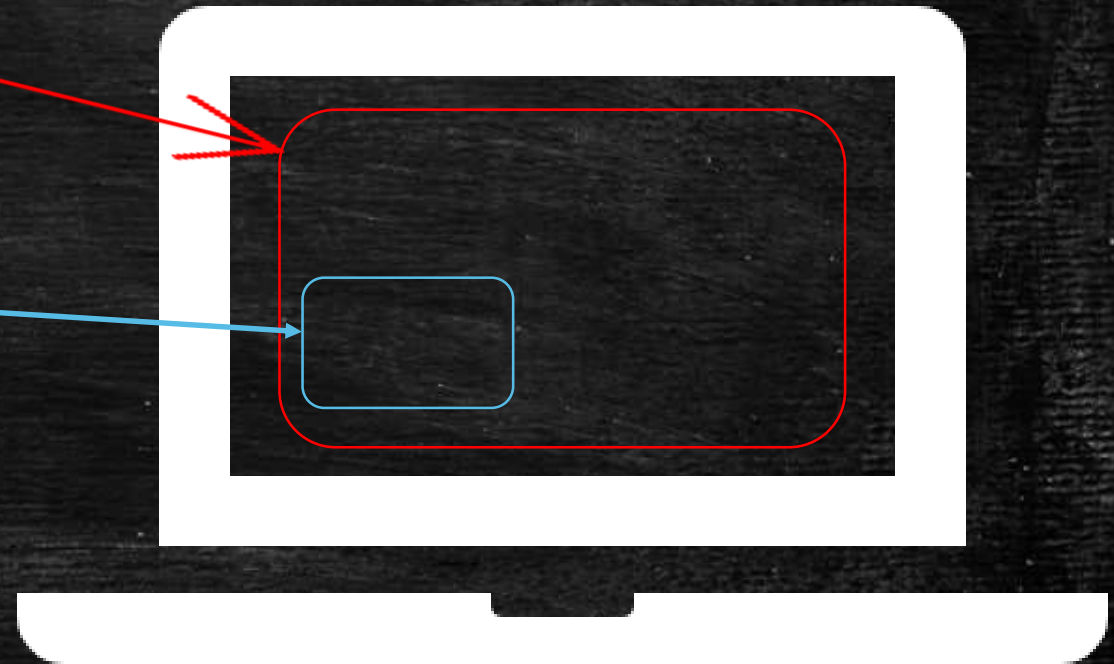
- Make non real space for your device
- For providing many Operating systems on the same Device or hardware
- Make Many servers on the same server

Virtualization



Windows

linux



Virtualization

- Virtual machine can share the same ip from network or can use another ip from the main ip
- Assume Windows is 192.168.1.5
- If you used bridge mode then Linux would be 192.68.1.6 or 192.168.1.x
- But if you used NAT then you can get ip from the main ip as
- 192.168.45.12

Questions

